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Palaeomagnetic Measurements on Karroo Dolerites

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§ 1. Introduction

In 1955 the authors started on a programme of measurement of the direction of magnetization of the sedimentary rocks of the Karroo system. In selecting specimens for palaeomagnetic measurement it is necessary to avoid regions of dolerite intrusion in view of the possibility of the sediments being baked by such intrusions. It is well known that there are extensive dolerite intrusions in the Karroo system north and east of the line marked on the 1939 edition of du Toit's map (du Toit 1939) as the southern limit of the Karroo dolerites, but omitted from the 1954 edition. Dolerites of possible Karroo age do occur south of this line but comparatively infrequently. However, in this southern region the Karroo sediments have been affected by the Cape folding and in some cases show evidence of shear. Thus, these rocks too, must be used with caution for palaeomagnetic measurements.

In view of the dolerite difficulty it became clear that it was desirable to measure the direction of magnetization associated with the dolerites at an early stage. The first collection of 30 dolerite samples from 13 sills and dykes exposed in road cuttings and quarries was made in December 1955 during a visit of Dr. John W. Graham of the Department of Terrestrial Magnetism, Carnegie Institution of Washington. On a later trip the authors collected a further 17 samples from 7 similar fresh exposures and six additional samples from a sill sampled during the first trip. The results of this survey were disappointing in that they showed considerable scatter. Remeasurement after a period of seven or eight months showed that the scatter could be considerably reduced by the elimination of the samples which were demonstrably unstable. Details of the measurements of the surface samples will be given in § 6 of this paper.

In the meantime measurements had been made on underground samples from two localities viz : (*a*) from the shafts of Winkelhaak Gold Mine to the east of Johannesburg and (*b*) from a railway tunnel near Estcourt in Natal. The scatter in the underground samples was considerably less than that in the surface samples as had been found by Gough (1956) in a similar survey of the Pilansberg dyke system.

§ 2. Method of Collection

In general the sample collected is in the form of a one inch diameter cylinder some five or six inches long. This is drilled *in situ* using a portable diamond coreing drill driven by a small petrol engine. The principal difficulty in drilling away from road transport is that a minimum

of two gallons of water is required to drill a six inch core. A line is scratched down the length of the core while still *in situ* and the bearing of this line and the inclination of the axis of the cylindrical core in two directions are determined by means of a Brunton compass. The top and bottom of the core are carefully marked.

In the first collection of dolerites oriented hand samples were chipped from the rock and drilled in the laboratory. A similar procedure was used by the staff of the Geology Department of Union Corporation in collecting samples from the shafts of Winkelhaak mine.

Usually four or five specimens in the form of cylinders one inch in diameter and one inch long are cut from each sample whether it is drilled *in situ* or in the laboratory.

§ 3. Method of Measurement

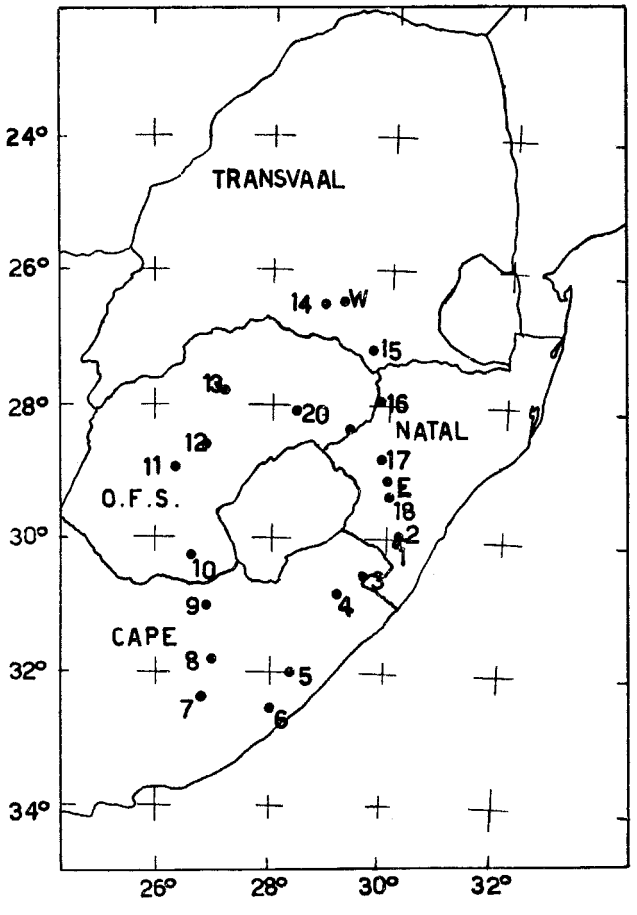
The instrument used for measurement is a spinner type magnetometer similar to that described by Graham (1955). In this type of magnetometer a signal from the coils, generated by the spinning specimen, is compared in phase with that received by a photoelectric cell from a paint pattern on the spinning top. In the system as devised by Graham the signals from the coils and the photoelectric unit are fed through carefully matched narrow band pass filters before they are compared in a phase discriminating circuit. For the early measurements reported here a circuit involving only one filter unit was used. The signals from the coil and the photoelectric unit were mixed before filtering. If the signal from the photoelectric unit be adjusted until it is equal in amplitude but opposite in phase to that from the coils, then the resultant signal received from the filter will be a minimum. The filter unit in this case functions as a null detector. This method works very well down to intensities of magnetization of the order of 10^{-6} c.g.s.u. giving an accuracy at this level of better than 2 or 3 degrees. However, as the signal approaches noise level, the accuracy with which it is possible to set to a minimum falls rapidly so that the system containing two separate filters, as used by Graham, is very much more convenient for measuring weakly magnetized specimens. The Department of Terrestrial Magnetism, Carnegie Institution of Washington, has very kindly lent us a pair of their matched filter units and the instrument was modified to operate in the Graham manner before measurements were made on the specimens from the Winkelhaak shafts and the Estcourt tunnel.

§ 4. Measurements of Underground Dolerites from Winkelhaak, near Kinross, Transvaal

Oriented samples were very kindly collected for us by geologists of Union Corporation during the course of shaft sinking operations at Winkelhaak Mines Ltd. on the Far East Rand. The position of the mine is marked as 'W' on the map of the Eastern half of the Union shown in fig. 1. Two pairs of vertical shafts were sunk, the members of a pair being approximately 750 feet apart and the distance between the pairs about one mile.

Two dolerite sills intruded into Lower Karroo (Ecca) sediments, were intersected in the shafts. The upper sill extends from the surface to a depth of between 70 and 80 feet. The lower sill, occurring at a depth of about 630 feet, is about 50 feet thick. Seventeen samples, subsequently cut into 57 specimens, were collected from the upper sill, while 8 samples, cut into 33 specimens came from the lower sill.

Fig. 1

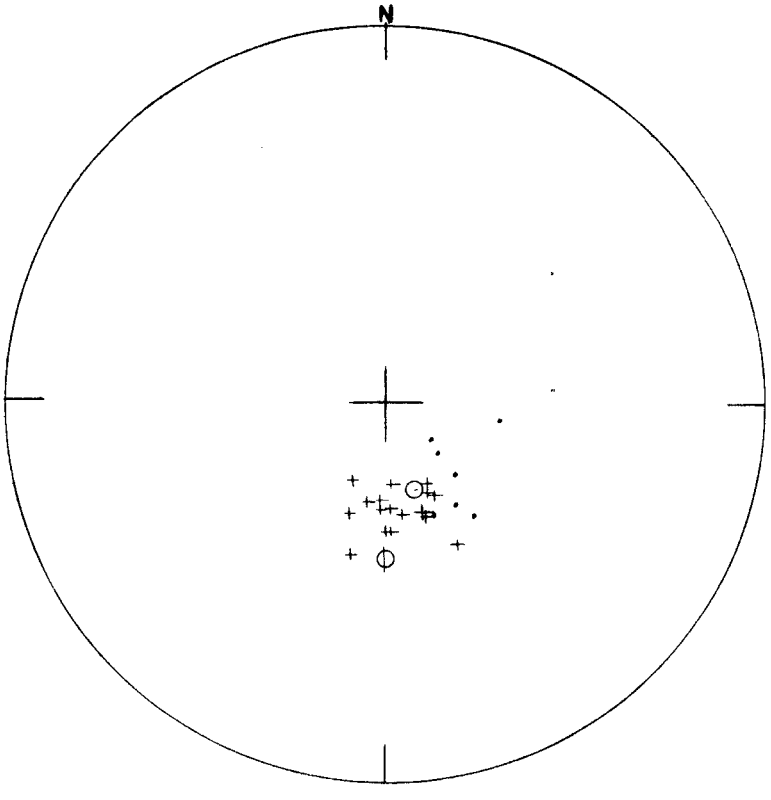


The Eastern half of the Union of South Africa. The numbered points indicate the localities at which surface samples were collected. 'E' and 'W' indicate Estcourt and Winkelhaak respectively.

All specimens from the upper sill were magnetized with the north seeking poles downwards, i.e. in the reverse direction to the present field, whereas all specimens from the lower sill had the south seeking pole downwards, i.e. are magnetized in the normal sense. The results of these measurements are shown in fig. 2, a stereographic plot of the lower hemisphere. It should be noted that the directions found for the upper and

lower sills are not exactly reversed relative to each other. The directions of magnetization of the samples from the lower sill have steeper dips and lie farther east than those of the upper sill.

Fig. 2

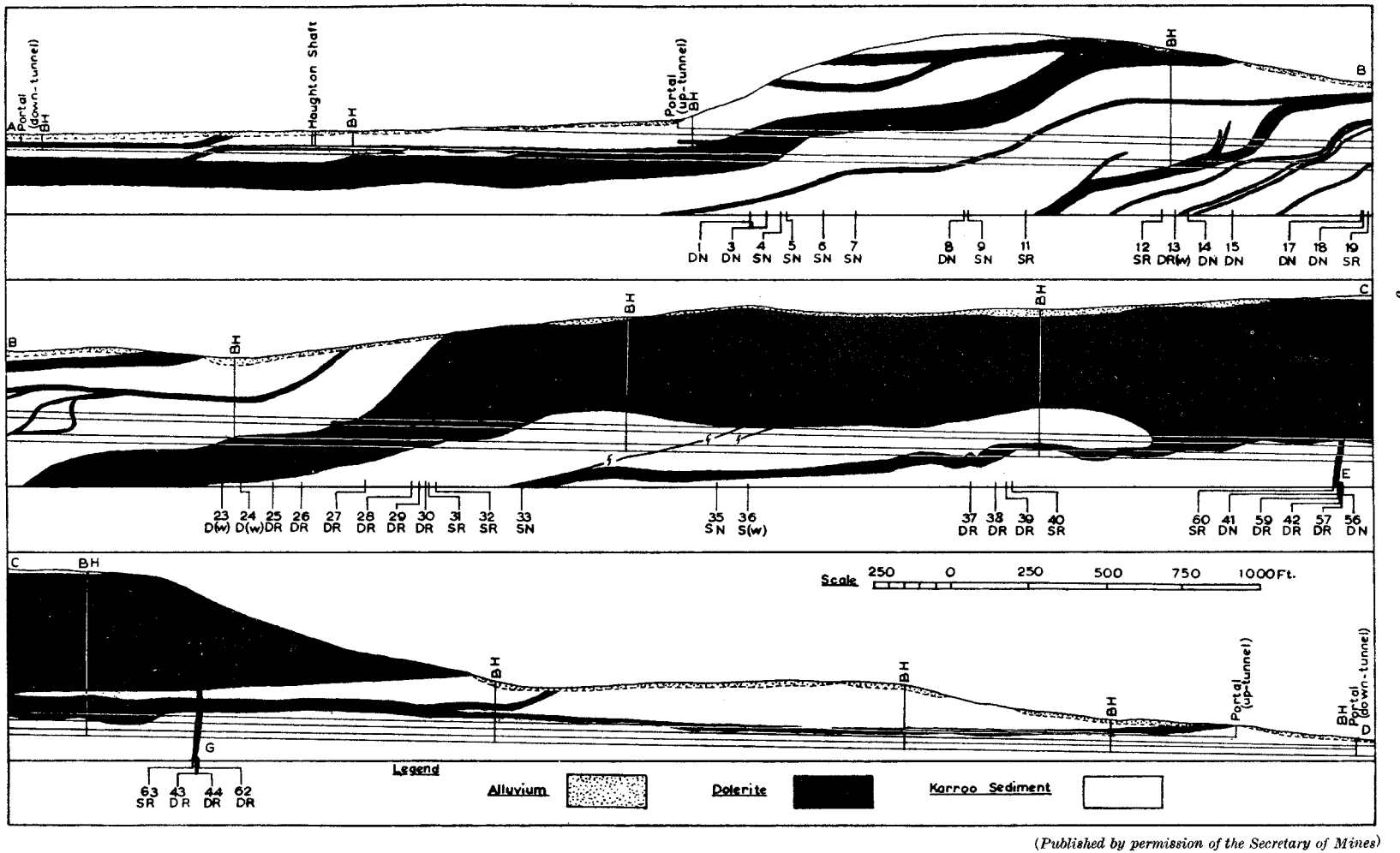


Directions of magnetization of the dolerite samples from Winkelhaak plotted on the lower hemisphere of a stereographic net.

• South-seeking poles (lower sill).
+ North-seeking poles (upper sill).
○ Present field.
⊕ Axial dipole field.

The statistics for these samples are given in table 1. In calculating the mean directions, sample means were first calculated from the directions of the individual specimens. These sample means were then used to calculate the mean for the group, giving each unit weight. If the measurements on each specimen had been given unit weight the circle of 95% confidence would have been much smaller but it is doubtful whether directions found from several specimens cut from the same sample can be regarded as independent measurements in terms of sampling the dyke or sill as a whole.

Fig. 3



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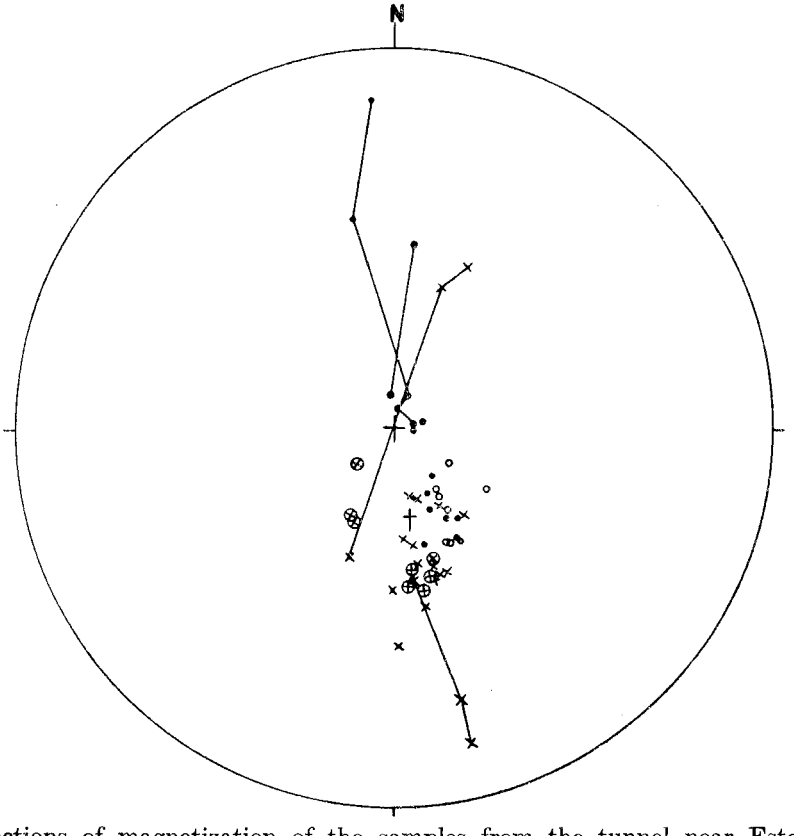
A geological section of the environs of the tunnels near Estcourt. The samples were taken from the lower tunnel vertically above the points indicated by numbers.

D=Dolerites ; S=Baked sediments ; N=Normally magnetized ; R=Reversely magnetized ; (W)=Scattered.

§ 5. Measurements of Underground Samples from a Railway Tunnel near Estcourt, Natal

A geological section of the tunnel environs, prepared by Mr. P. J. Smit of the Geological Survey, is shown in fig. 3. The samples were taken from the lower tunnel vertically above the points indicated by numbers on the diagram. Dolerite and baked sediment samples are distinguished by

Fig. 4



Directions of magnetization of the samples from the tunnel near Estcourt plotted stereographically on the lower hemisphere. Individual specimens from a single sample showing a streak are joined by a line. Otherwise sample means are plotted.

• =Normally magnetized dolerites; × = Reversely magnetized dolerites; ○=Normally magnetized baked sediments; ⊕=Reversely magnetized baked sediments; † =Direction of the present field of the earth.

' D ' and ' S ' respectively while ' N ' indicates that they were magnetized in the normal sense and ' R ' in the reversed sense. In the case of four samples (13, 23, 24, 36) the individual specimens from a single sample form a streak instead of a fairly tight group. These are indicated by ' W '. The directions of magnetization are plotted on the lower hemisphere of a stereographic net in fig. 4. For the four samples showing the

streaking the directions of magnetization of the individual specimens have been plotted, those from the same sample being linked by a line. The rest are shown as the mean for the sample, calculated from several specimens.

Examination of the data in fig. 3 shows that in general the baked sediments are magnetized in the same sense as the adjacent dolerites. There are, however, some anomalies.

When the first collection was made two samples (41, 42) were taken from the dyke at point E. One of these (41) was found to be magnetized in the normal sense, the other in the reversed sense. It was decided to return to the tunnel to collect additional samples from this dyke. (The other anomalous regions had already been cemented up.) These samples served to show that the chill zones were normally magnetized, whereas the interior of the dyke and the only sample of baked sediment that was measurable, were magnetized in the reversed sense. Possible explanations of this are :

(*a*) that the chill zones are unstable ;

(*b*) that a self-reversing mechanism operates in the chill zones ;

(*c*) that a self-reversing mechanism operates in both the interior of the dyke and in the adjacent sediments but not in the chill zones ;

(*d*) that a reversal of the earth's field occurred while the dyke was cooling ;

(*e*) that there is in fact more than one intrusion with a reversal of the earth's field between the intrusions.

The possibilities (*c*), (*d*) and (*e*) seem unlikely. On the information available at present it is impossible to decide between (*a*) and (*b*), though the possibility of instability in some samples of dolerite is supported by the streaking noted in samples 13, 23 and 24.

Table 1

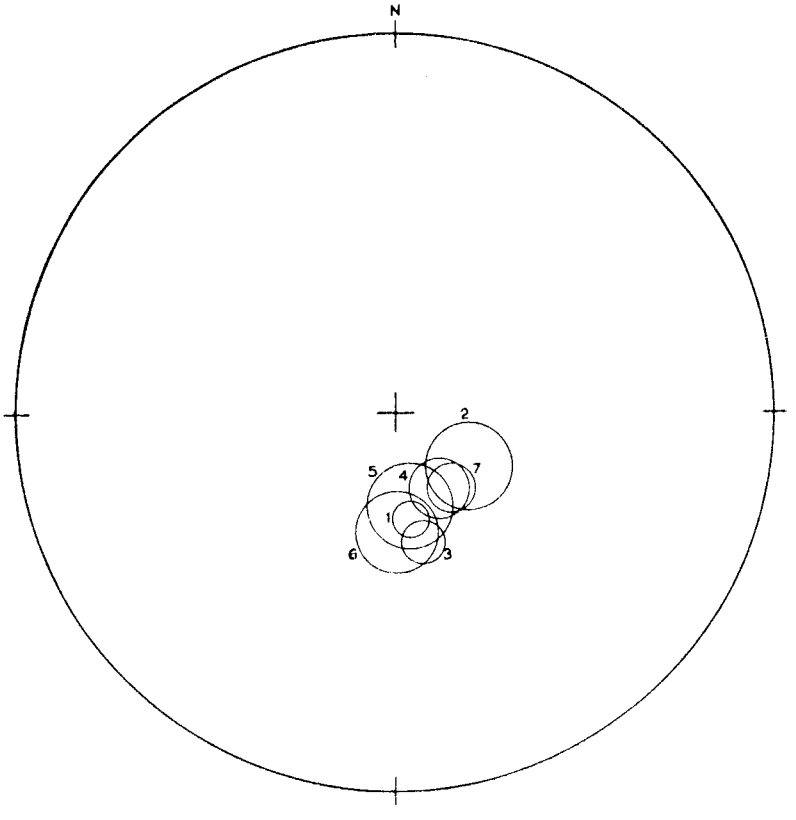
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Latitude of collection point	26·5°S	26·5°S	29·1°S	29·1°S	29·1°S	29·1°S	30·3°S
Longitude of collection point	29·1°E	29·1°E	29·9°E	29·9°E	29·9°E	29·9°E	28·5°E
Number of samples	17	8	15	8	9	7	33
Number of specimens	57	33	53	25	34	24	149
Mean declination, D	173°E	127°E	167°E	180°E	151°E	144°E	172°E
Mean dip, I	58°	63°	51°	55°	64°	62°	62°
Semi-vertical angle of 95% confidence, α †	5°	12°	6°	11°	8°	7°	12°
Geographic pole	77°S	44°S	77°S	—	67°S	—	76°S
Geographic longitude of pole	53°E	82°E	90°E	—	94°E	—	52°E
δp	6°	14°	6°	—	10°	—	14°
δm	7°	18°	8°	—	12°	—	19°

(1) Winkelhaak upper sill ; (2) Winkelhaak lower sill ; (3) Estcourt reversed dolerites ; (4) Estcourt reversed sediments ; (5) Estcourt normal dolerites ; (6) Estcourt normal sediments ; (7) Surface collection.

† Calculated by Fisher's Method (1953).

Considering the results from this tunnel as a whole, mean directions calculated from the reversely magnetized dolerites, the reversely magnetized sediments, the normally magnetized dolerites and the normally magnetized sediments are given in table 1. Samples 13, 23, 24 and 36 have been omitted from these calculations. The difference between the sediments and the dolerites is in all cases within the limits of error of the means, whereas the mean for the normally magnetized samples is

Fig. 5



Stereographic projection on the lower hemisphere showing the circles of 95% confidence of the mean values for (1) Winkelhaak reversed dolerites, (2) Winkelhaak normal dolerites, (3) Estcourt reversed dolerites, (4) Estcourt normal dolerites, (5) Surface dolerites, (6) Estcourt reversed baked sediments, (7) Estcourt normal baked sediments.

significantly different from that of the reversely magnetized group. This point is clearly shown in fig. 5, a lower hemisphere stereographic plot showing the 95% confidence circles for the four groups of specimens from Estcourt and the two groups from Winkelhaak. It will be noted also that the Winkelhaak reversed samples do not differ significantly from the

Estcourt reversed samples nor do the Winkelhaak normal samples differ significantly from the normal Estcourt samples. All the normally magnetized samples do, however, differ significantly from the reversed ones. It may be expected that, in view of the possible instability of some of the normally magnetized samples, the mean for these samples might not be as satisfactory a datum as that found from the reversely magnetized ones. In the case of the latter it is considered that the fact that they have retained a direction of magnetization which is in the opposite sense to the present field is in itself an assurance of their stability.

Inspection of fig. 4 reveals that, although the normally magnetized samples give magnetic directions closer to the present field of the earth than do the reversed ones, they all lie off to the East, and it seems unlikely that they are all unstable. Furthermore, it is difficult to see why, in general, the baked sediments should only be unstable close to dolerites that are unstable. Similarly, it seems unlikely that a self-reversal mechanism should affect dolerites and their baked sediments in some areas while those in other areas should be unchanged.

The explanation of these observations would seem to be complex. It seems probable that these intrusions are not strictly contemporaneous and the possibility of a reversal of the earth's field during the time of intrusion of this swarm of sills and dykes must be considered. Perhaps samples 13, 23 and 24 happened to be cooling through their Curie points at the time of the reversal of the field. If we assume that such a reversal did take place, we would have a picture of normally magnetized sills and dykes with normally magnetized adjacent sediments and reversely magnetized dolerites with their reversed baked sediments. Secondary effects, due to self-reversal under special circumstances or to instability, might be superimposed on the simple picture.

§ 6. The Measurements on the Surface Samples

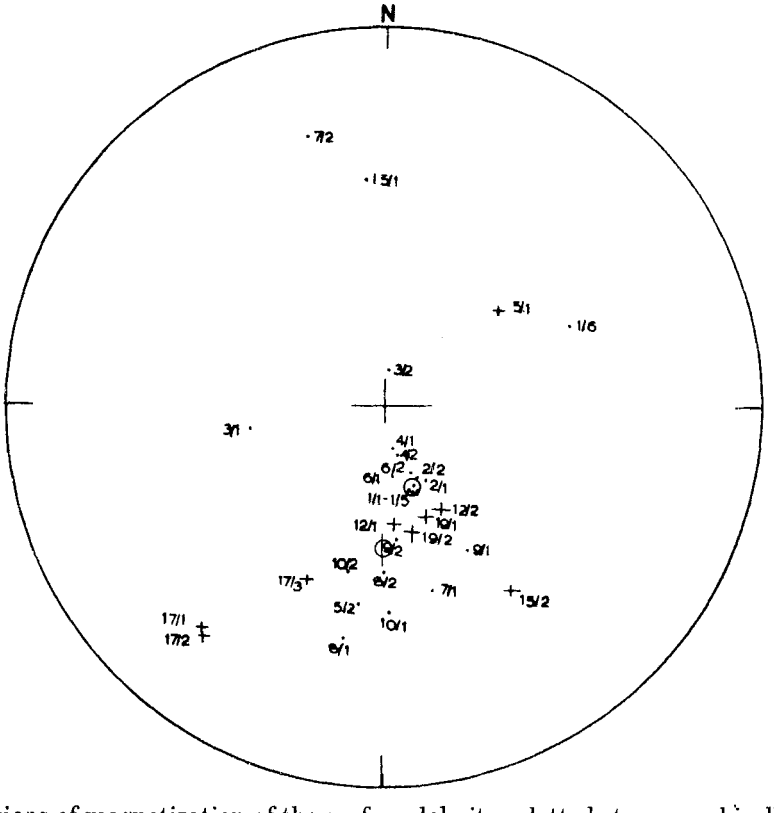
The localities from which the samples were collected are indicated by numbers on a map in fig. 1. All the surface samples were measured as soon after collection as possible. One specimen from each sample was remeasured three or four months later and another specimen from each was remeasured some seven or eight months after the first measurements. It was found that some specimens showed a considerable change from the direction of magnetization first measured and it seems reasonable to conclude that those specimens for which the change was considerable are unstable. It was decided to exclude all samples from a locality if any one of the specimens remeasured had shown a change in the direction of magnetization of more than 10 degrees. The sample means for the remaining samples are shown in fig. 6. It will be noted that there are still a number of scattered measurements but in such cases the two samples from the same locality do not agree and they, too, may be unstable. The mean for the surface collection was calculated disregarding sign and giving unit weight to each sample and is given in table 1. As will be seen

from a plot of this mean and its circle of 95% confidence (fig. 5) the surface samples are not inconsistent with the Winkelhaak or Estcourt groups.

§ 7. Discussion

All samples discussed in this paper were taken from intrusions in sediments of Karroo age, or from the sediments themselves. There is no doubt that the intrusions can be classified as Karroo dolerites. These dolerites have been studied extensively by Walker and Poldervaart (1949).

Fig. 6



Directions of magnetization of the surface dolerites plotted stereographically on the lower hemisphere.

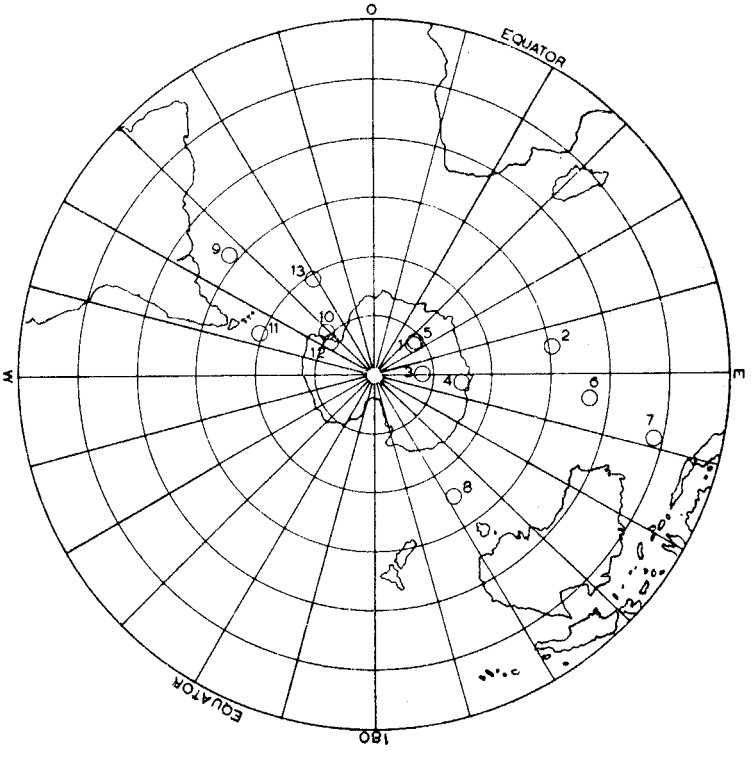
• = Normal sense ; + = Reversed sense ; ⊙ = Present field ; ⊕ = axial dipole field.

(17/2 indicates the mean direction of magnetisation of specimens from sample 2 taken from locality 17.)

They consider that the Stormberg lavas were poured out towards the close of Triassic time and that the dolerites were more or less contemporaneous with the lavas. Du Toit (1954) remarks also that there is an intimate connection between the Stormberg basalts and the dolerites and concludes that the dolerites appear to have been intruded at the commencement of the Jurassic epoch.

The positions of the magnetic poles corresponding to the mean directions of magnetization found in §§ 4, 5 and 6 have been calculated on the assumption that the field at the time was a dipole field. These values are given in table 1 together with quantities δp, δm, the semivertical angles of the oval cone of 95% confidence (Irving 1956 a). These positions have been plotted in fig. 7, an azimuthal equidistant projection of the southern hemisphere. Also plotted in fig. 7 are pole positions inferred in similar manner from measurements on specimens from England, North America, India and Tasmania. Details of ages and authorities are given in table 2.

Fig. 7



An azimuthal equidistant projection of the southern hemisphere showing the poles calculated from palaeomagnetic data from various parts of the world. (See table 2.)

It will be seen that the pole positions inferred from the measurements on the South African dolerites are not consistent with any of those found for rocks of the same age in Europe, North America and Australia. If it be assumed that the direction of magnetization, when averaged over the time of intrusion of these dolerites, is a slowly and smoothly varying function of time then the simplest way of reconciling the divergent pole positions

is to postulate relative displacement of the continents, i.e. Continental drift. It is noteworthy that the displacements required are of the kind proposed by Wegener (1924), du Toit (1937), King (1953) and Carey (1955).

It should be noted, however, that there is some evidence that the assumption of a slowly varying direction of magnetization is an over-simplification. Clegg *et al.* (1956), point out that the lavas from the Khandala region are younger than those from Linga. Since the Khandala measurements give a pole position nearer to India than do the Linga

Table 2

1. Winkelhaak upper sill—Jurassic.
2. Winkelhaak lower sill—Jurassic.
3. Estcourt, reverse sense—Jurassic.
4. Estcourt, normal sense—Jurassic.
5. Surface collection—Jurassic.
6. Linga area, late Cretaceous to early Eocene, Clegg *et al.* (1956).
7. Khandala area, early Eocene, Clegg *et al.* (1956).
8. Tasmanian dolerite, probably Jurassic, Irving (1956 b).
9. Keuper marls, Triassic, Clegg *et al.* (1954).
10. Antrim lavas, Eocene, Hospers and Charlesworth (1954).
11. Arizona, Triassic, Runcorn (1955).
12. Colorado plateau, Cretaceous, Runcorn (1955).
13. Maryland, Triassic, J. W. Graham (1955).

measurements, this means that either the drift pattern was complicated, or that there was polar movement of some 20 degrees between the times of intrusion of the two series of lava flows. It is interesting to note that the difference in the direction of magnetization of the reversed and normal South African dolerites is indicative of a fairly rapid movement of the pole or of the continent.

It is clear that more accurately dated measurements covering a larger proportion of the stratigraphic column than are at present available are required in order to establish with certainty whether or not the present misfit of palaeomagnetic data is due to imperfect correlation. Only then will one be able to describe with any confidence the possible drift paths followed by the continents.

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